

DIRECT: A Deterministic Lightweight Safe Vision Control Framework for Flexible Cable Insertion

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20 Hz

Closed-loop servoing, full camera rate

84

Annotated frames to train perception

68.54

Avg. SFP trial score (/100)

52.33

Avg. SC trial score (/100)

ZERO

Force & off-limit contact penalties, 30 trials

DIRECT Path

Trajectory efficiency on every trial

1. ABSTRACT

Cable insertion is a dexterous manipulation task that remains a persistent bottleneck in industrial electronics assembly. The Intrinsic ‘AI for Industry’ Challenge formalises it as a competitive sim-to-real testbed demanding **sub-millimetre precision under tight force constraints**. End-to-end Vision-Language-Action (VLA) models promise automation, but introduce unquantified latency, massive data requirements and edge-case failures that are unacceptable in high-stakes manufacturing.

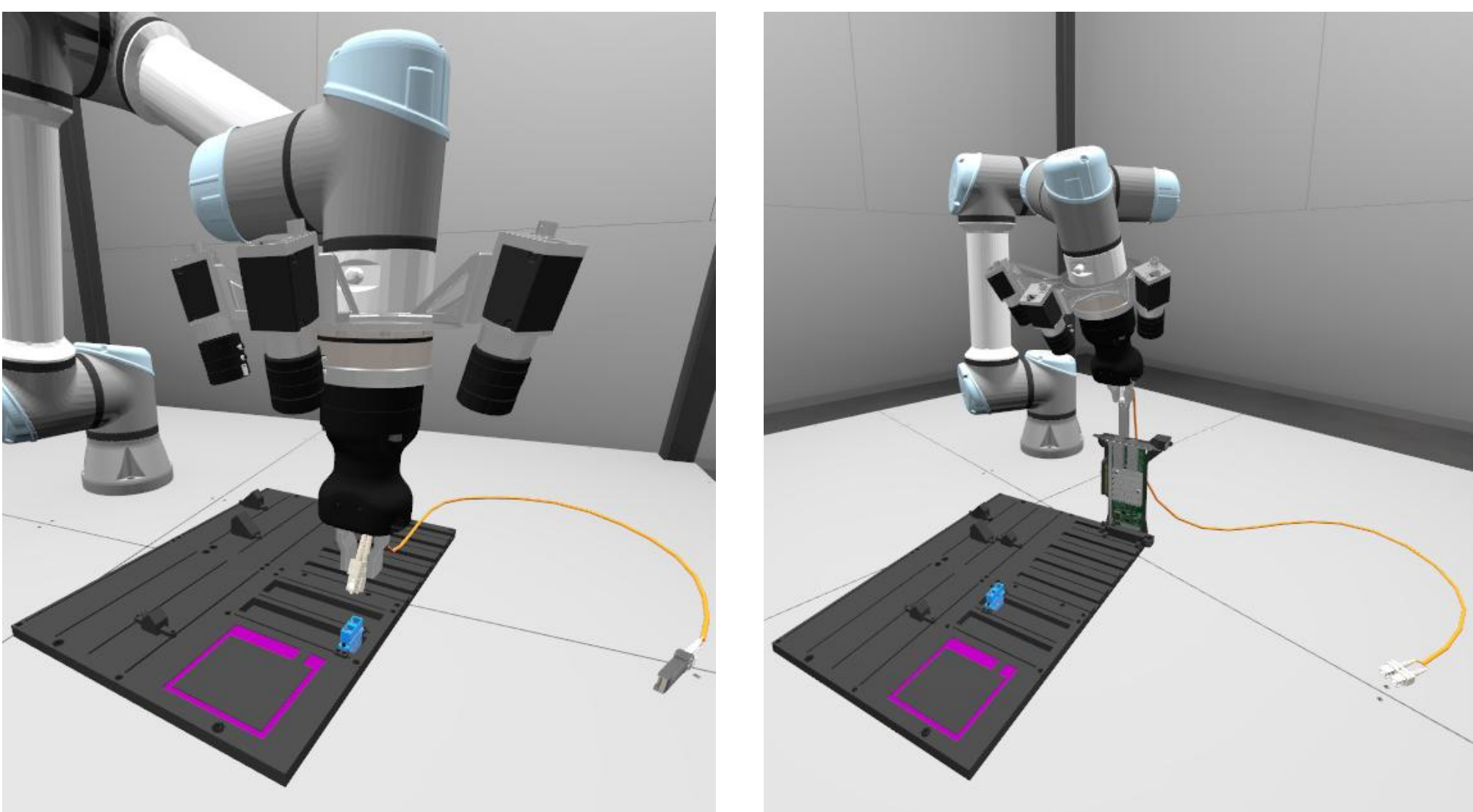
We present **DIRECT** — Deterministic Insertion via Real-time Estimation and Control Tracking — a decoupled pipeline that deliberately avoids end-to-end generative action models, pairing real-time detector-based socket localisation from a wrist-mounted camera with a deterministic, admittance-controlled insertion planner.

Index Terms — Robotic Manipulation · Cable Insertion · Object Detection · Trajectory Planning · Industrial Automation

2. KEY CONTRIBUTIONS

- A deterministic, fully interpretable perception–estimation–actuation pipeline for flexible cable insertion, extensible to generic single-arm peg-in-hole tasks.
- YOLOv26 instance segmentation for real-time port detection, with bounding-box-area depth estimation that eliminates the need for 3D cameras.
- A connector-specific insertion strategy: each cable type is mapped to pre-specified roll, Y-travel and Z-push sequences that apply auditable contact forces.
- Validation on the Intrinsic AI for Industry benchmark with high trajectory efficiency and zero safety-policy violations across 30 trials.
- Learning is confined to perception — control stays deterministic, sidestepping the sim-to-real gap in the contact phase.

3. SIMULATION ENVIRONMENT



(a) SFP module insertion into NIC card slot.

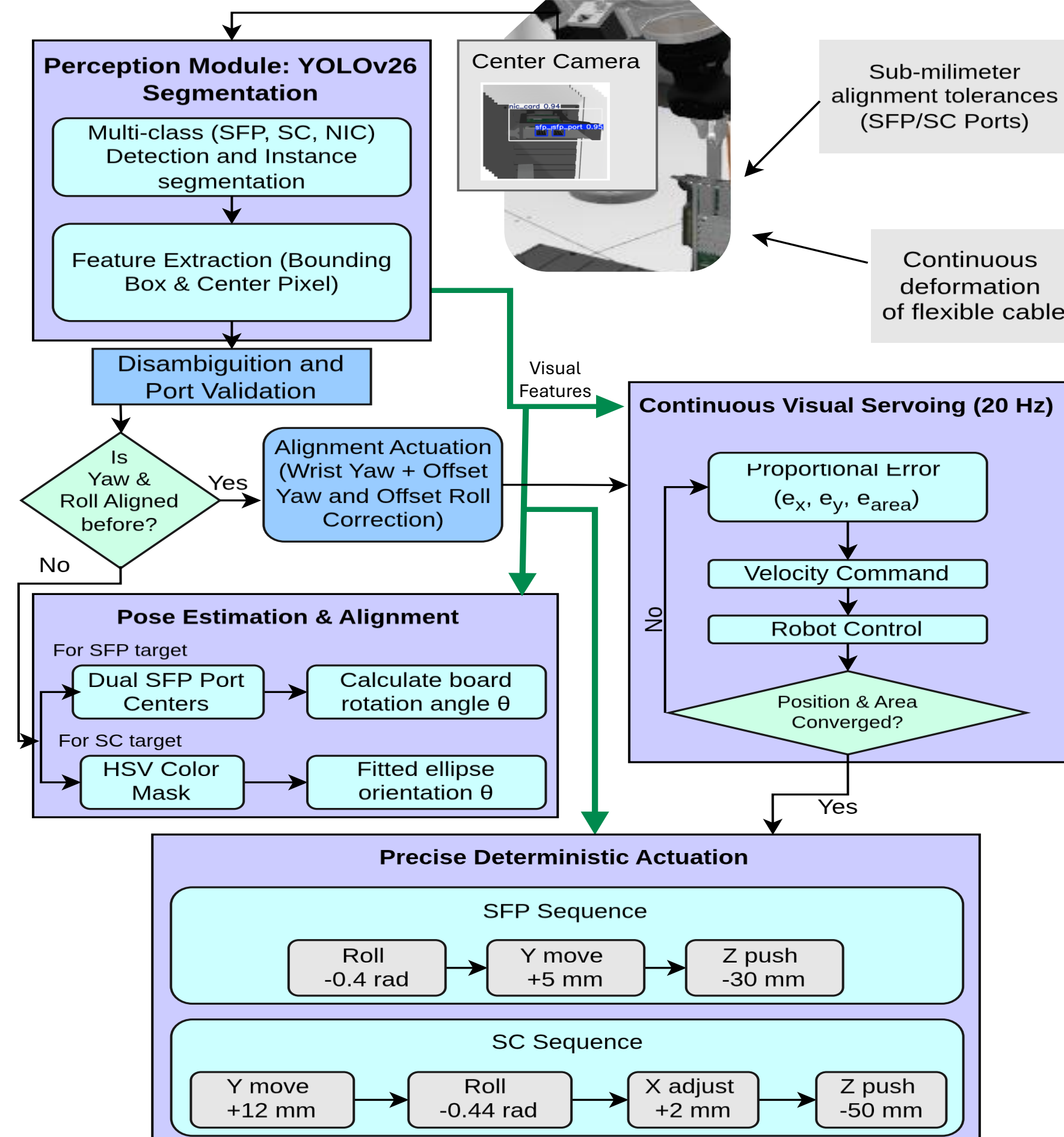
(b) SC plug insertion into optical port.

Sample Gazebo trial screenshots from DIRECT pipeline execution. Left: SFP insertion with 4-panel debug grid showing detection overlays and error vectors. Right: SC insertion with cyan port segmentation.

4. EVALUATION ENVIRONMENT AND SCORING

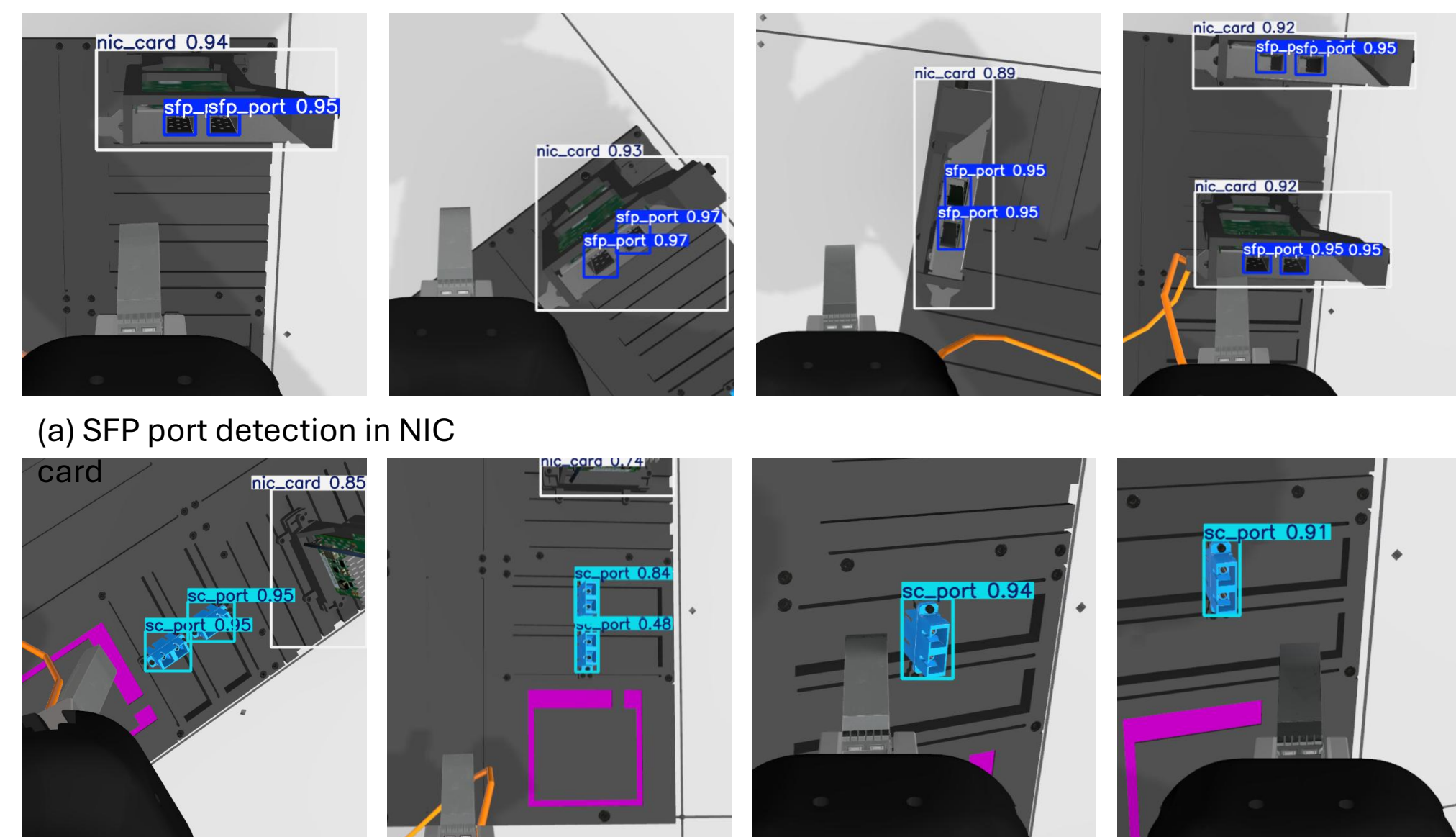
- Environment.** Gazebo (Jetty) simulation of the Intrinsic AI for Industry task board, orchestrated over ROS 2 by the `aic_engine`, which spawns the board, runs the trial and collects scores; the F/T sensor is tared before every run.
- Randomised high-mix board.** Board pose and yaw, NIC-card rails and SC port positions are randomised per trial, so the robot relies entirely on perception, not fixed pick locations.
- Tiered scoring, 100 pts / trial.** Tier 1 validity (1) + Tier 2 smoothness 6, duration 12, efficiency 6, with -12 force and -24 off-limit penalties + Tier 3 success (75 full, 38–50 partial, 0–25 proximity); cumulated over Trials 1–2 (SFP) and Trial 3 (SC).

5. THE DIRECT FRAMEWORK



High-level diagram of DIRECT Framework (Deterministic Insertion via Real-time Estimation and Control Tracking)

6. PORT DETECTION — YOLOv26



Sample object detection using trained YOLOv26

7. PERCEPTION & CONTROL

Detection. yolo26l-seg trained on just 84 hand-annotated Gazebo frames with data augmentation across three classes (sfp_port, sc_port, nic_card).

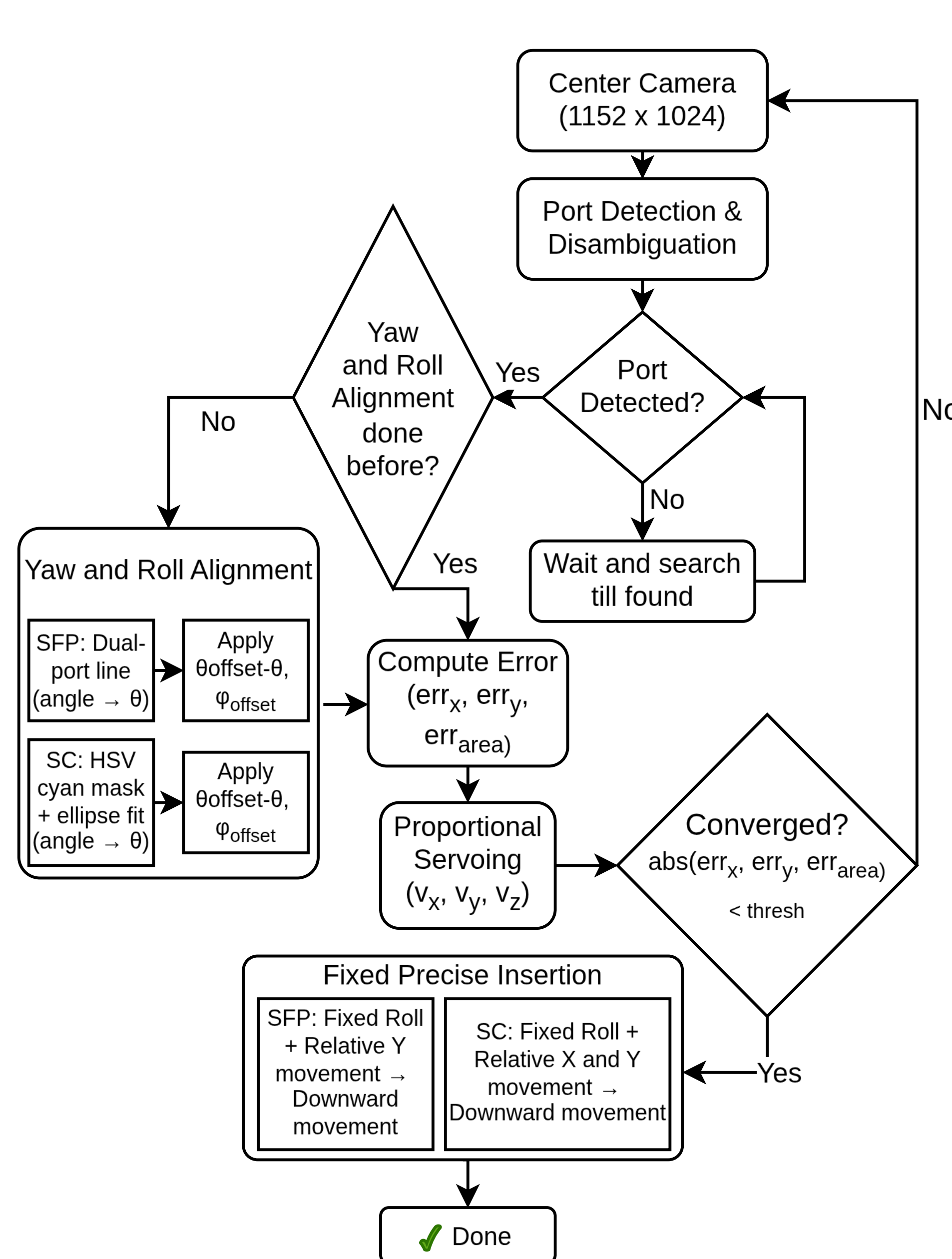
Yaw alignment. Board angle $\theta = \arctan(\Delta y / \Delta x)$ from dual SFP centres; HSV cyan mask + rotated-rect orientation for SC ports.

Servoing. $v_x = K_p(\alpha_x - \alpha_x^*)$, $v_y = -K_p(c_y - c_y^*)$, $v_z = K_d(A - A^*)$; converges at ≤ 1 px centre error and ≤ 20 px² area error.

Insertion. SFP: roll -0.40 rad $\rightarrow$ +5 mm Y $\rightarrow$ -30 mm Z. SC: +12 mm Y $\rightarrow$ roll -0.44 rad $\rightarrow$ +2 mm X $\rightarrow$ -50 mm Z.

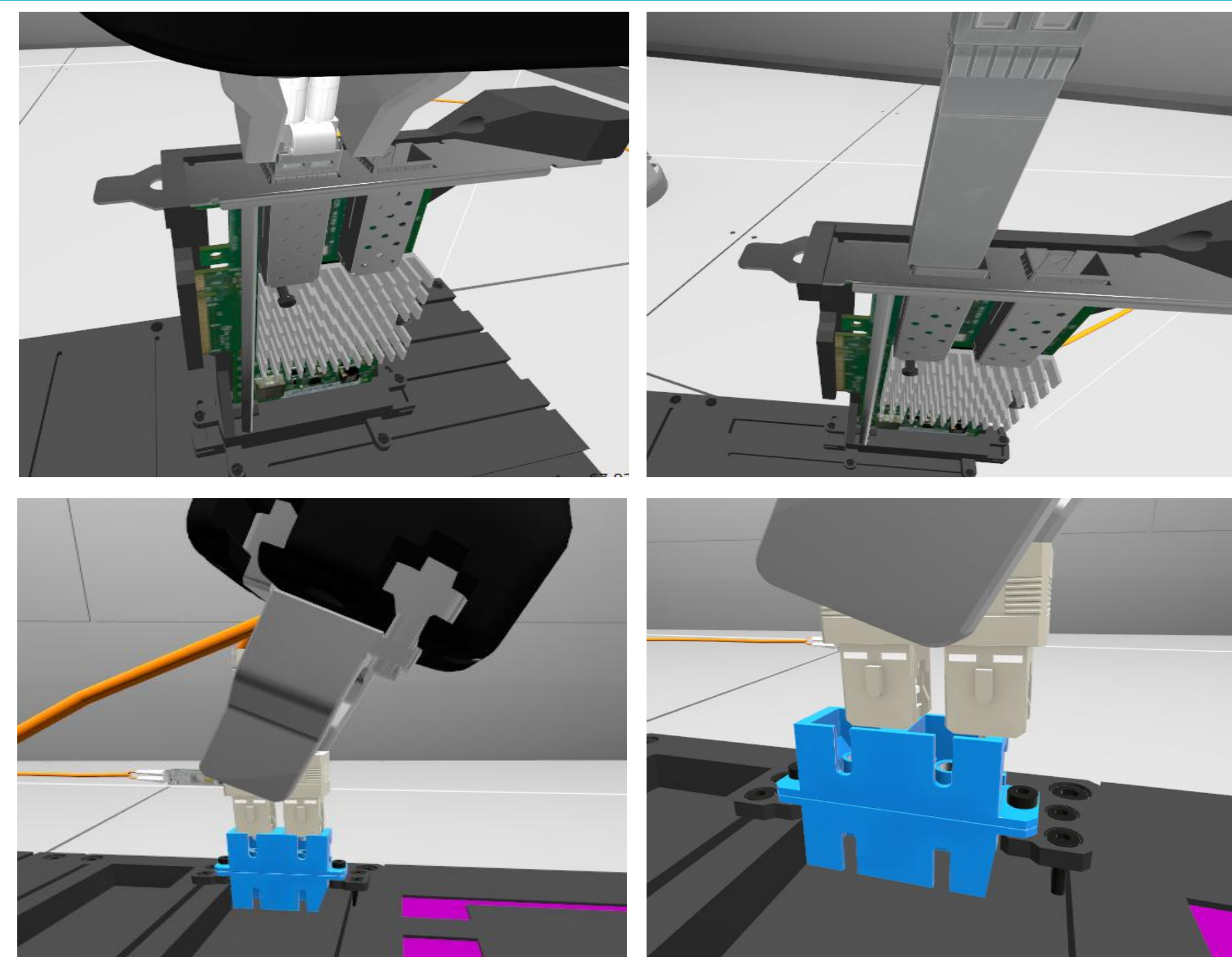
Platform. UR5e + Robotiq Hand-E, ATI AXIA80-M20 F/T sensor, 3× Basler acA2440-20gc (1152×1024, 20 FPS), ROS 2 / Gazebo.

8. DIRECT FLOWCHART



High-level DIRECT flowchart: YOLOv26l-seg detection followed by proportional visual servoing with connector specific hardcoded insertion sequence.

9. GAZEBO TRIALS — INSERTION SUCCESS & FAILURES



Sample success and failure cases of SFP and SC plug insertion screenshots in Gazebo Sim

10. TRIAL SCORE BREAKDOWN

TABLE I: Sample Trial Scores and breakdown by AIC Engine for a single run

Metric	Trial 1 (SFP)	Trial 2 (SFP)	Trial 3 (SC)
Tier 1 Score	1	1	1
Tier 2 Score	12.141	13.041	16.393
Contacts [†]	0	0	0
Duration (Time)	0.615 (57.18 s)	1.507 (53.09 s)	5.224 (36.06 s)
Insertion Force [†]	0	0	0
Traj. Efficiency (end-effector path len., initial plug-port dist.)	6 (0.00 m, 0.13 m)	6 (0.00 m, 0.14 m)	6 (0.00 m, 0.32 m)
Traj. Smoothness (Avg. jerk)	5.526 (3.95 m/s ³)	5.534 (3.88 m/s ³)	5.169 (6.93 m/s ³)
Tier 3 Score	24.52	75.00	39.59
Total (183.68)	37.66	89.04	56.98

[†] Penalty scores.

11. AGGREGATE SCORES — 10 RUNS

TABLE II: Scores obtained by AIC Engine for total ten runs

Runs	Scores (Total = Tier 1 + Tier 2 + Tier 3)		
	Trial 1 (SFP)	Trial 2 (SFP)	Trial 3 (SC)
Run 1	38.01 (=1+12.158+24.85)	89.03 (=1+13.03+75)	57.08 (=1+16.563+39.52)
Run 2	37.66 (=1+12.141+24.52)	89.04 (=1+13.041+75)	56.98 (=1+16.393+39.59)
Run 3	61.41 (=1+13.001+47.41)	38.14 (=1+12.26+24.88)	56.69 (=1+16.184+39.51)
Run 4	37.99 (=1+12.332+24.66)	89.44 (=1+13.442+75)	1 (=1+0+0)
Run 5	37.84 (=1+12.137+24.70)	89.41 (=1+13.405+75)	57.52 (=1+16.11+40.41)
Run 6	37.89 (=1+12.109+24.78)	88.53 (=1+12.532+75)	56.42 (=1+15.912+39.51)
Run 7	90.73 (=1+14.729+75)	88.87 (=1+12.866+75)	57.11 (=1+16.598+39.51)
Run 8	61.48 (=1+13.071+47.41)	89.63 (=1+13.635+75)	56.67 (=1+16.16+39.51)
Run 9	37.79 (=1+12.136+24.65)	89.04 (=1+13.055+75)	56.72 (=1+16.125+39.6)
Run 10	89.56 (=1+13.558+75)	89.21 (=1+13.213+75)	67.11 (=1+16.397+49.71)
Average	53.04	84.03	52.33

12. CONCLUSION

DIRECT demonstrates that industrial reliability, interpretability and safety can be achieved without end-to-end learned policies. Across 30 trials the pipeline incurred no force or off-limit contact penalties, required no safety intervention, and damaged no hardware — while running at the full camera frame rate with no GPU inference in the control loop.

Deterministic modular pipelines remain a competitive alternative to VLA systems in structured, safety-critical industrial environments. Remaining headroom lies in cycle time (avg. 55.4 s), which duration-weighted scoring penalises — not in safety or accuracy.